
GIT PUSH THE BEATS

Melody Dive Use-Case Specification: Searching

Version 1.1

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Use-Case Specification: Searching	Date: 15/11/2025
UC Searching	

Revision History

Date	Version	Description	Author
15/11/2025	1.0	Add: add details information about this use case	Thái Minh Huy
21/11/2025	1.1	Add: Add Semantic Search in Alternative Subflow	Thái Minh Huy
21/11/2025	1.2	Modify: Modify Brief Description	Thái Minh Huy

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Use-Case Specification: Searching

1. Use-Case Name: Searching

1.1 Brief Description

This use case allows the user to find specific songs within the system by entering keywords or applying filters on Genre and Composer. Beside, the user can have the option to search the music by using natural language queries. The user can interact with the results by playing music or adding songs to their favorites or add them into their personal playlists.

2. Flow of Events

2.1 Basic Flow

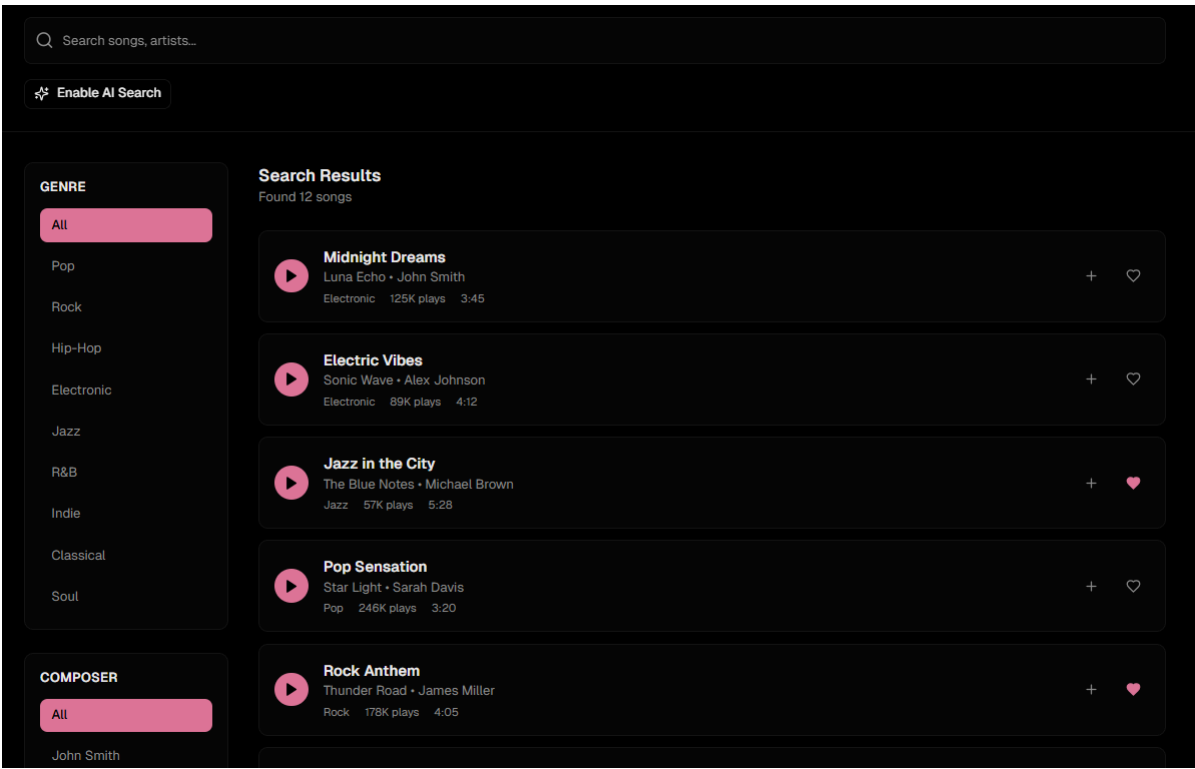


Figure 1: The interface of Searching

1. The system displays the Search Interface, which includes a top search bar with the placeholder “Search songs, artists...” and a sidebar filter categories (Genre, Composer).
2. The system displays a list of default or random songs in the “Search Results” area, along with a counter indicating the number of matches (Example: “Found 12 songs”).
3. The user clicks into the search bar and enters a keyword (Example: a song title or artist name).
4. The system updates the “Search Results” list in real-time (or upon pressing Enter) to show only songs matching the input.

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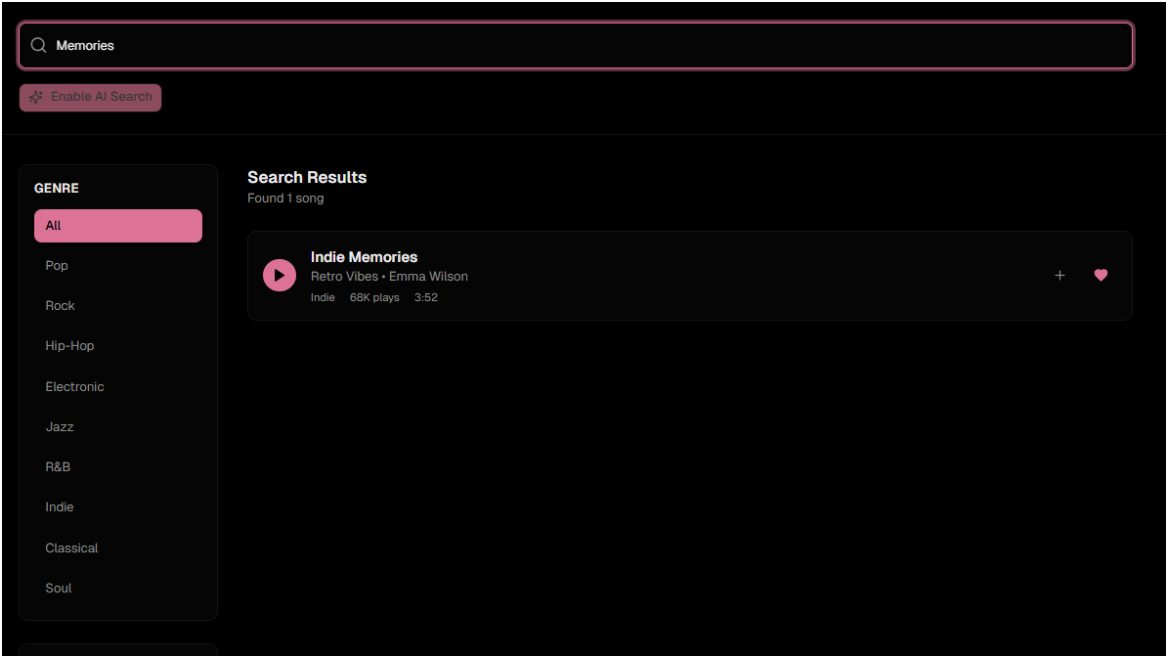


Figure 2: Interface searching with results

5. The system displays the detailed metadata for each result, including:
 - Play Button: An icon with a “Play” triangle.
 - Song Info: Title, Artist/Composer.
 - Stats: Genre, Play Count and Duration.
 - Favorite Status: A heart icon indicating if the song is liked.
 - Add button: Add a song to the user's personal playlists.
6. The user browses the results.

2.2 Alternative Flows

2.2.1 Filter by Category (Genre/Composer)

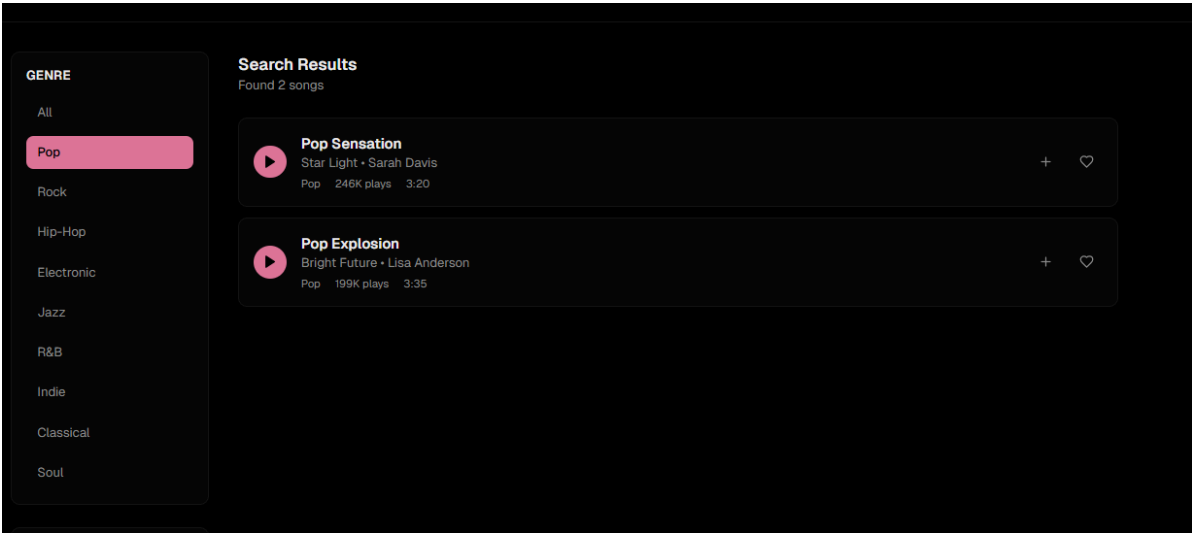


Figure 3: Filter interface

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- At step 3 of the Basic Flow, instead of typing text, the user can select a specific filter from the sidebar (Example: Select “Jazz” by genre or “John Smith” under Composer).
- The system highlights the selected filter (Example: “All” is currently highlighted in the image).
- The system filters the “Search Results” list to display only items matching that category.

2.2.2 No results found

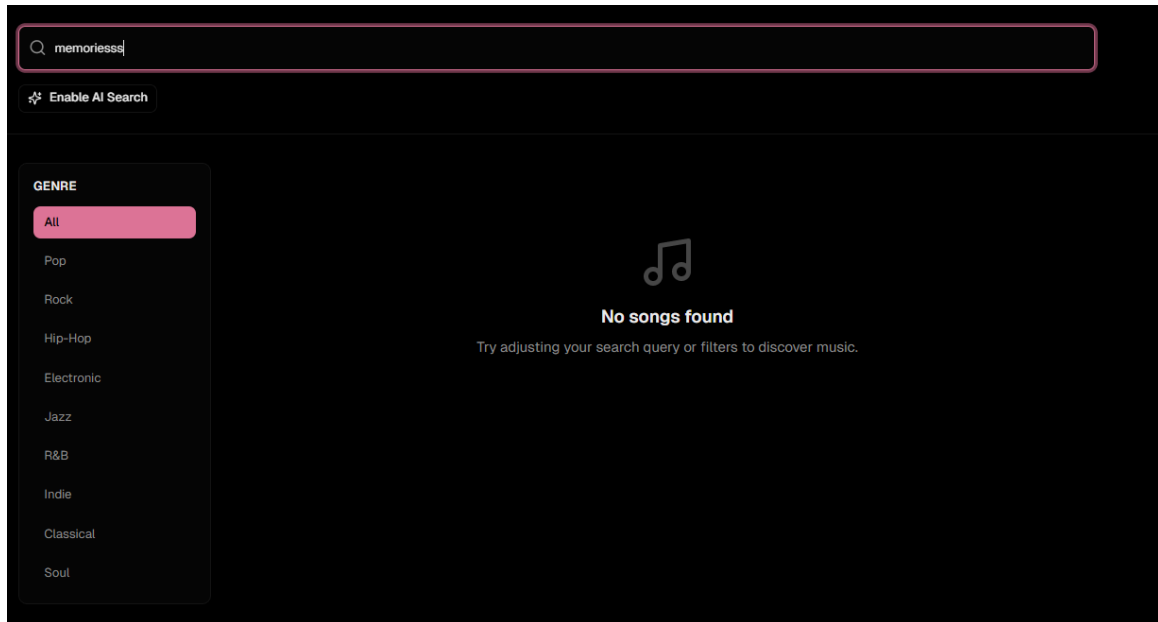


Figure 4: Interface of searching with no results

If the search keywords or filters yield no matches:

- The system updates the counter to “No songs found”.
- The system displays a message indicating no results were found.

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2.2.3 Semantic Search

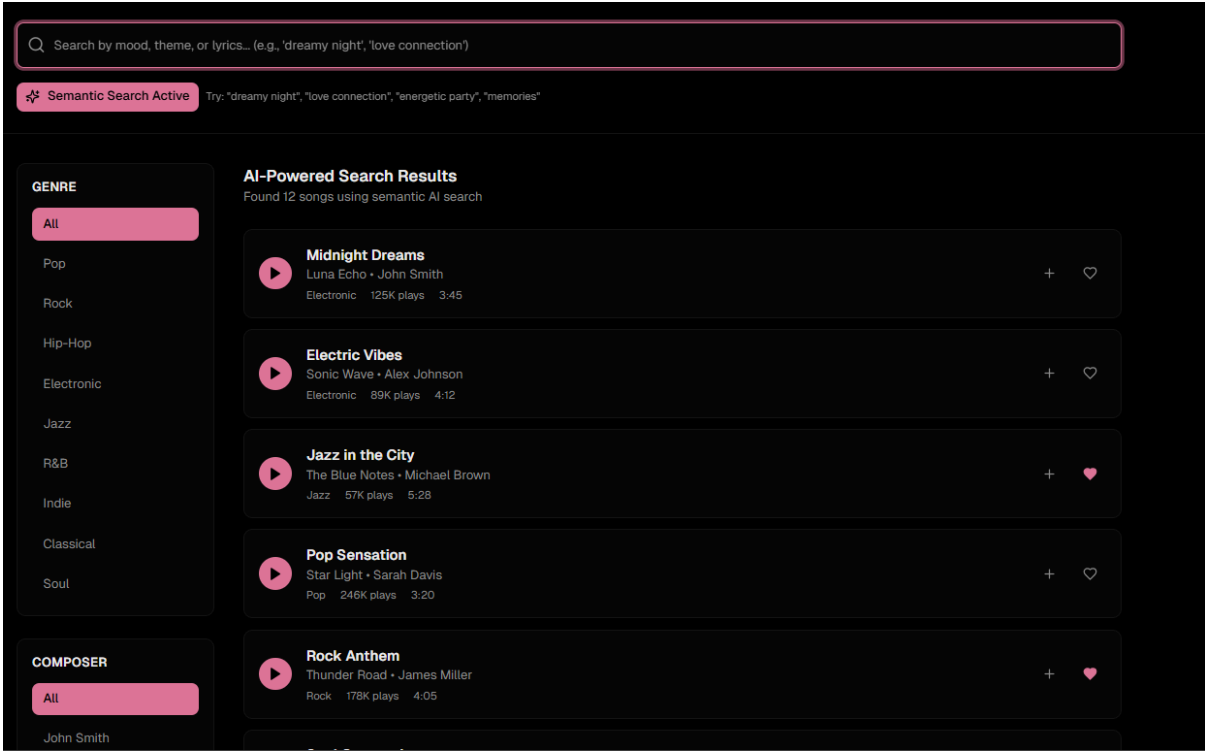


Figure 5: The interface of semantic search

1. The system displays the Search Interface with the “Semantic Search Active” indicator visible and highlighted in pink color.
2. The system provides a search input field with placeholder text guiding the user: “Search by mood, theme, or lyrics...(‘dreamy night’, ‘lovely connection’).
3. The system may display suggested semantic queries below the active indicator (Example: try: 'dreamy night', 'love connection', 'energetic party', 'memories').
4. The user enters a natural language phrase describing the desired vibe or theme (Example: “energetic party”) into the search bar.

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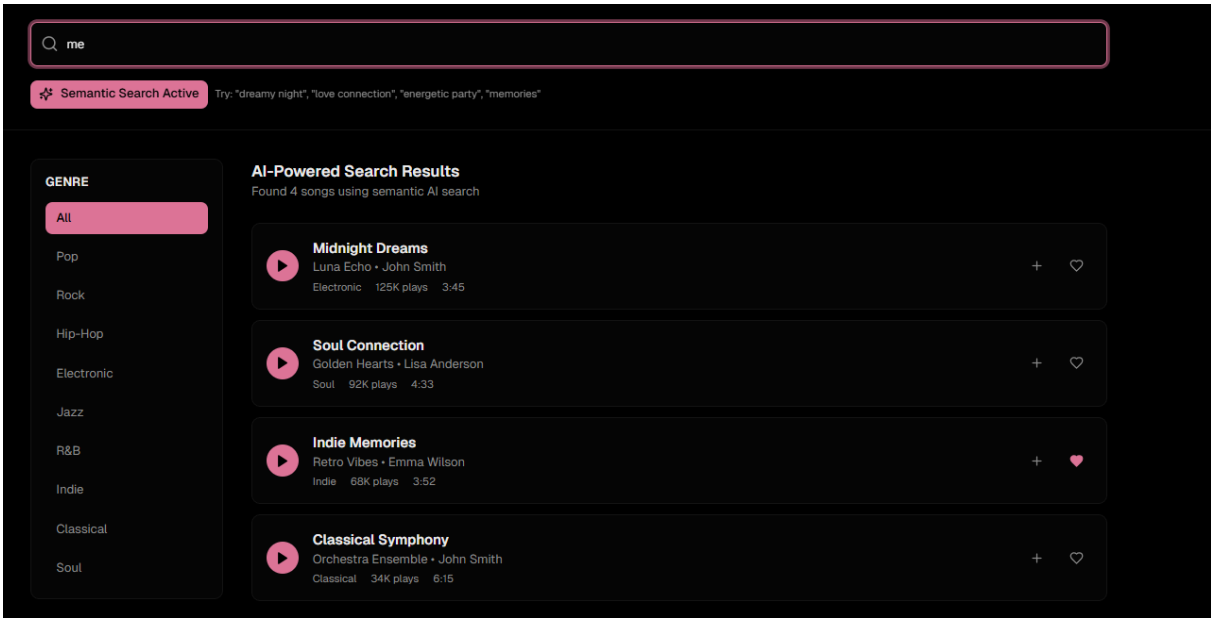


Figure 6: Interface of search bar in semantic search

5. The system processes the input using an AI semantic engine to understand the context and intent.
6. The system updates the results section, displaying the header “AI-Powered Search Results” and the count of matches.
7. The system lists the songs that semantically match the query. Each result includes:
 - Play Button
 - Metadata
 - Action: Add to playlist button and Favorite Buttons
8. The user interacts with the results (Play, Add or Favorite).

2.2.3.1 No Semantic Matches Found

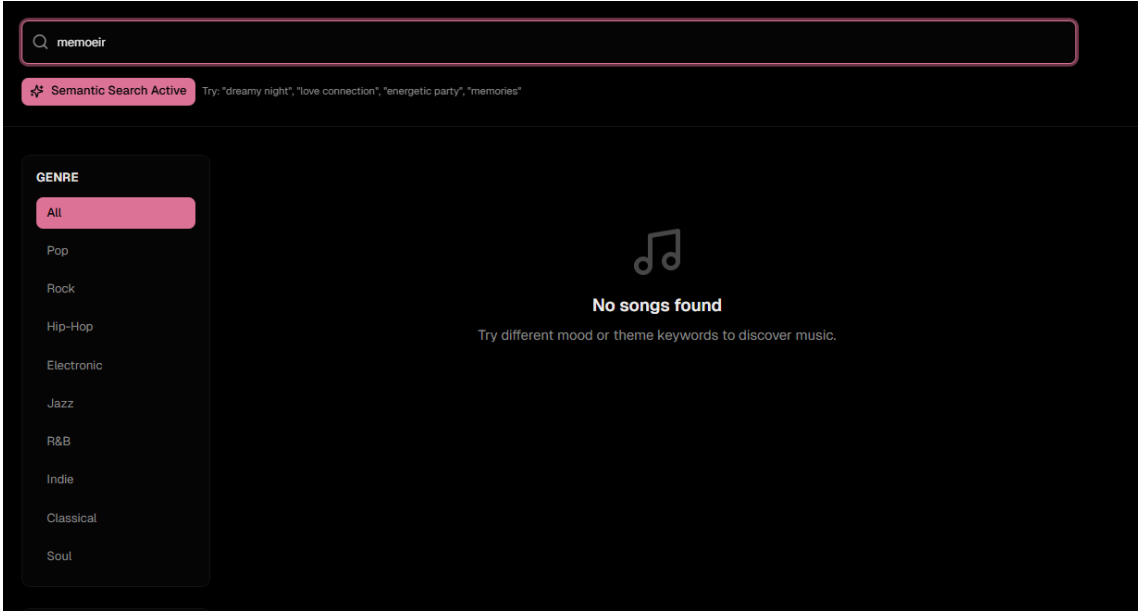


Figure 7: The interface of no result in semantic search

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If the AI engine cannot interpret the user’s input or finds no songs with matching themes:

- The system displays “No Songs found”.
- The system suggests using other keywords to find.

2.2.3.2 Filtering AI Result

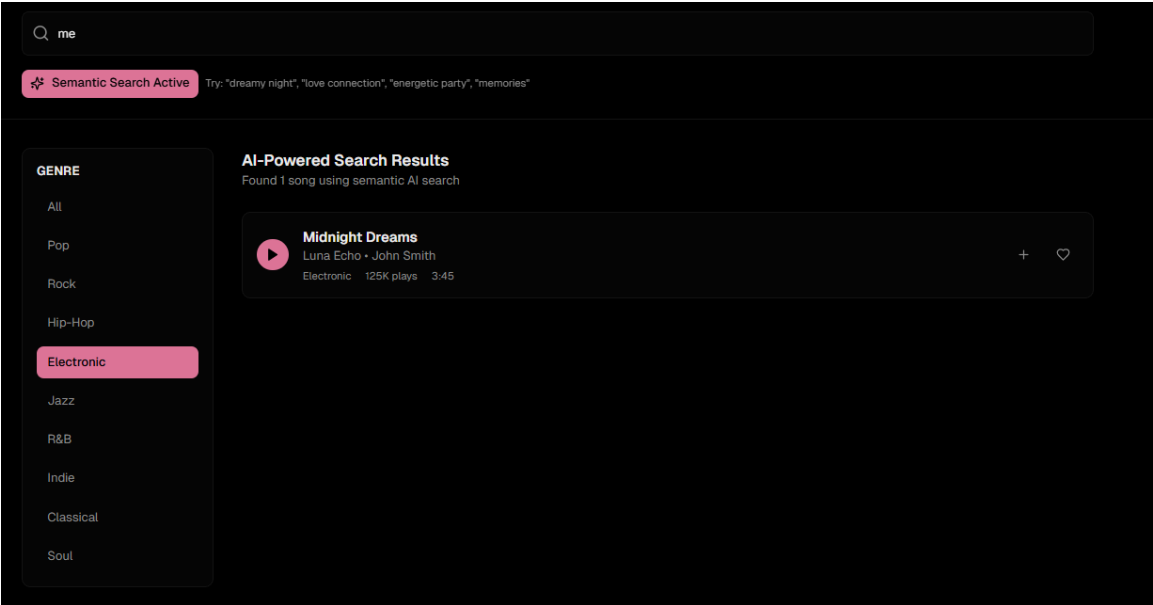


Figure 8: Interface of semantic search with filter

After step 7, the user selects a filter from the Genre or Composer sidebar. The system refines the existing AI-generated list to show only songs that match both the semantic query and the selected genre.

3. Special Requirements

3.1 Visual Feedback

Selected filters must be visually distinct from unselected ones (Example: highlighted in pink for selected).

3.2 Play Count Formatting

Large numbers for play counts must be abbreviated for readability (example: 125K instead of 125,000).

3.3 AI Feedback Indicators

The interface must tell the user the difference between Normal Search Mode and Semantic Search Mode.

4. Preconditions

4.1 AI Service Online

The backend AI/NLP service must be operational to process natural language queries (AI Features).

4.2 Featured Enabled

The user must click the “Enable AI Search” Button and change it to “Semantic Search Active” status to activate A.I Search.

5. Postconditions

5.1 Display State

The user sees a curated list of songs matching their specific criteria.

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5.2 Interaction

The user can immediately play a song or toggle its favorite status from this view.

6. Extension Points

6.1 Toggle Favorite

Location: Step 5 of basic Flow or Step 7 of Alternative flows 2.2.3.

Description: The user clicks the heart icon next to a song.

- If the heart is outlined, the system fills it in pink and adds the song to the user's favorite.
- If the heart is filled, the system reverts it to an outline and removes the song from favorites.

6.2 Play Song

Location: Step 5 of basic Flow or Step 7 of Alternative flows 2.2.3.

Description: the user clicks the Play button on a specific row. The system begins audio playback for that track.

6.3 Toggle to Playlist

Location: Step 5 of basic Flow or Step 7 of Alternative flows 2.2.3

Description: The user clicks the plus symbol next to a heart icon.

- The system will display to users "Which playlist do you want to store in a playlist" and show all personal user playlist.
- The user will select one song and store it to multiple playlists (One song can be in zero or many in playlists).
- After that, the song will now be stored in the playlist and back to the basic flow of step 5 and 6.